Two Attack Surfaces of a Shared Quantum Computer: Hardware Crosstalk and Circuit-Level Tampering

Subarna Adhikari, Zeeshan Ajmal, Juuso Herajärvi and Kimmo Halunen
OUSPG, University of Oulu, Finland

Why This Matters

For quantum computers (QCs) to be commercially viable, providers are likely to move towards multi-tenant user model where multiple users can access the system at the same time. Some QCs already execute circuits from different users on the quantum processor simultaneously. While this maximizes resource efficiency, it also creates a security problem that has no classical analogue: your computation now shares fragile qubit fabric with a stranger's. We study ways that this 'sharing' can be exploited and how it can be detected.

Threat Model

- Cloud-accessible, multi-tenant quantum processor.
- Adversary is an ordinary co-tenant: Same public interface, no privileged hardware access.
- Attack surfaces studied: Hardware-level interference between co-located qubits, and circuit-level tampering of a submitted program.

Hardware Crosstalk

Operations on neighboring qubits leak noise, degrading a co-located computation.

Experimental Setup
QC: IBM Brisbane (127-qubit superconducting)
Victim Circuit: Bernstein–Vazirani
Attacker Circuit: Queues of two-qubit gates placed at controlled distances and orientations.

What We Found

- Distance is effective in reducing impact of crosstalk, but saturates. Changing the placement of the attacker qubit from directly adjacent to one buffer qubit away recovers roughly **3–5%** victim fidelity. Further separation does not have a measurable impact.

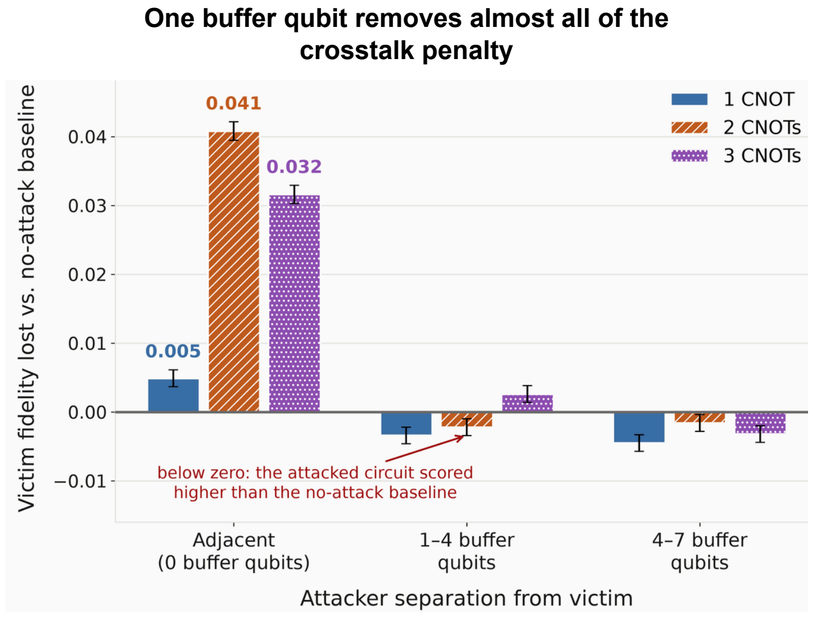


Figure1: IBM Brisbane, 127 qubits. Bernstein–Vazirani victim, N = 102,400 shots per point. Error bars ±1 s.e.

- Orientation matters. With a CNOT attacker adjacent to the victim, placing the target qubit next to the victim does more damage than placing the control qubit.
- Randomized sequences are the most damaging. Mixing different two qubit gate orientations with single-qubit operations decreased victim fidelity the most to **0.8359**.

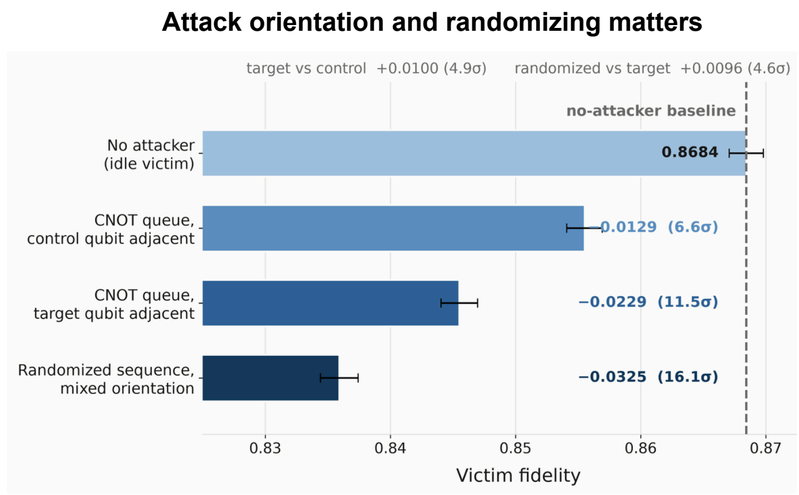


Figure2: IBM Brisbane, 127 qubits. Three idling victim qubits, attacker queue on adjacent qubits. N = 61,440 shots per condition. Error bars ±1 s.e. Axis truncated: the full range is 0 to 1.

Circuit Tampering Detection

An adversary can execute malicious circuits in the shared platform. Quantum Trojans, can corrupt results without any obvious trace by altering the circuit layout or inserting additional gates. Can such tampering be detected from the circuit itself?

Experimental Setup
Dataset: Clean and Trojan-injected versions of six canonical algorithms (Bernstein–Vazirani, Deutsch–Jozsa, Grover, Shor, QFT, QAOA). Generated in Qiskit.
Injected Behaviours: Unauthorised gate additions, entanglement distortion, decoherence-like manipulation.
Classification: Two classifiers compared on structural and probabilistic circuit features.

What We Found

- **96%** accuracy with Random Forest on the combined dataset.
- QSVM reached **88%** on the combined set, performing well on structurally regular circuits (QAOA, QFT, Shor) and poorly on Grover.
- Ranking features with RF matters for audit. The quantum kernel does not support this.
- Note: The evaluation set is small (24 circuits in the combined test split), so the results are indicative rather than conclusive.

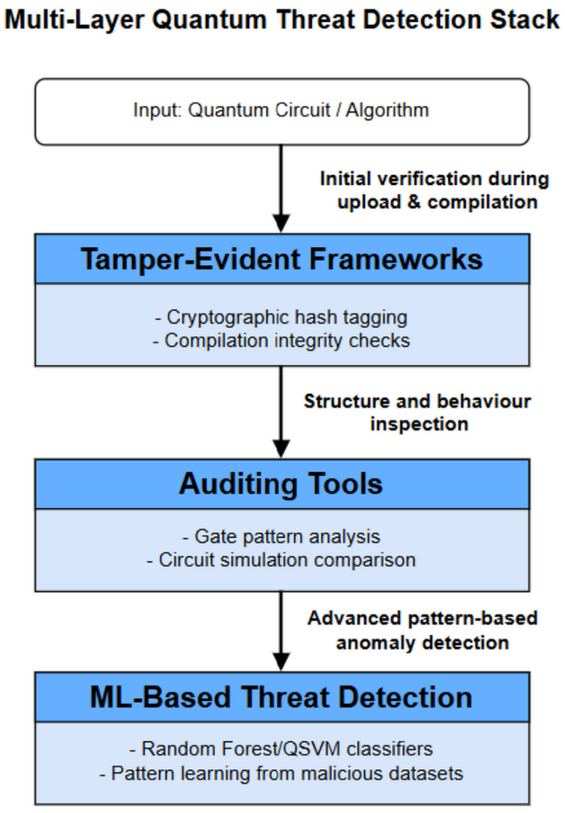


Figure3: Overview of the proposed multi-layered detection architecture.

To Summarize

Both attack frameworks exploit the same premise: a shared processor gives an adversary a channel to interfere with victim computation. Spatial isolation yields diminishing return beyond a specific threshold. Tampering Detection is useful in detection of known malicious patterns but struggles with novel variations and anomalies.

Where We Go Next

- Extend study of security issues accross different quantum architectures (Ion-trap, Photonic, Neutral atom)
- Expand the datset to ensure more robust and generalizable conclusions

References

A. Ash-Saki, M. Alam, and S. Ghosh, "Analysis of crosstalk in nisq devices and security implications in multi-programming regime," in Proceedings of the ACM/IEEE International Symposium on Low Power Electronics and Design, 2020, pp. 25–30. DOI: 10 . 1145/3370748.3406570.

N. Choudhury, C. N. Mude, S. Das, P. C. Tikkireddi, S. Tannu, and K. Basu. "Crosstalk-induced side channel threats in multi-tenant nisq computers." eprint: 2412.1 0507 (arXiv). [Online]. Available: https://arxiv.org/abs/2412.10507.

W. J. B. Lee, S. Wang, S. Dutta, W. E. Maouaki, and A. Chattopadhyay. "SWAP Attack: Stealthy Side-Channel Attack on Multi-Tenant Quantum Cloud System." eprint: 2502.10115 (arXiv). [Online]. Available:https://arxiv.org/abs/2502.10115.

S. Deshpande et al., "Design of quantum computer antivirus," in 2023 IEEE International Symposium on Hardware Oriented Security and Trust (HOST), 2023, pp. 260–270. DOI: 10.1109/HOST55118.2023.10133711.

S. Das and S. Ghosh, "Trojan taxonomy in quantum computing," in 2024 IEEE Computer Society Annual Symposium on VLSI (ISVLSI), 2024, pp. 644–649. DOI: 10.1109/ISVLSI61997.2024.00123

L.A. Wu and D. A. Lidar, "Quantum malware," Quantum Information Processing, vol. 5, no. 2, pp. 69–81, 2006. DOI: 10.1007/s11128-006-0014-5.

A. A. Arellano, H. Xie, and J. Szefer, "Qubitvise: Double-sided crosstalk attack in superconducting quantum computers," in 2025 IEEE International Conference on Quantum Computing and Engineering (QCE), IEEE,vol. 2, 2025, pp. 50–53.

Science with Arctic Attitude